

Advancing Surveillance Systems with LRCN: A Study on Abnormal Activity Recognition

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Abstract— Abnormal activity detection holds significant importance in various domains, including surveillance, healthcare, and industrial monitoring, for ensuring safety and security. Leveraging deep learning techniques has emerged as a promising approach in this realm, owing to their capability to automatically learn discriminative features from raw data. In this paper, we propose a novel framework for abnormal activity detection utilizing Long Short-Term Memory (LSTM) networks with Convolutional Neural Networks (CNN), known as Long-term Recurrent Convolutional Networks (LRCN). The proposed architecture integrates the temporal dependencies captured by LSTM with the spatial information extracted by CNN, enabling the model to effectively discern abnormal activities from normal ones in video sequences. Furthermore, we introduce a comprehensive dataset specifically curated for abnormal activity detection tasks, facilitating thorough evaluation and benchmarking of different methods. Extensive experiments conducted on benchmark datasets demonstrate the efficacy of the proposed LRCN framework, outperforming existing methods in terms of detection accuracy and robustness. Our findings underscore the potential of deep learning-based approaches, particularly LRCN, in addressing real-world challenges associated with abnormal activity detection, thus paving the way for enhanced safety and security systems.

Keywords— Abnormal activity detection; Surveillance; Deep learning; Anomaly detection; Video analysis.

I. INTRODUCTION

Surveillance system expansion and the rising need for safety have highlighted the importance of detecting abnormal activities [12]. Identifying unusual behaviors in video data automatically is crucial for enhanced security in areas like public safety, healthcare, transportation, [3] and industrial monitoring. Past methods for abnormal activity detection often use manual features or rule-based systems, which may struggle in diverse and intricate situations.

Deep learning methods have become a notable approach for abnormal activity detection by autonomously extracting meaningful features from data. [8] Convolutional Neural Networks (CNNs) are adept at grasping spatial details in images, and Recurrent Neural Networks (RNNs), especially Long Short-Term Memory (LSTM) networks, [1] excel at understanding temporal relationships in sequential data like videos.

Motivated by the strengths of CNNs and LSTMs, this paper proposes a novel framework for abnormal activity detection, termed [14] Long-term Recurrent Convolutional Networks (LRCN). By integrating CNNs for spatial feature extraction and LSTMs for temporal modeling, LRCN offers a holistic approach to analyze video data for abnormal activity detection. [10] The fusion of these two architectures allows

the model to effectively capture both short-term and long-term dependencies, enabling it to discern subtle abnormalities amidst complex and dynamic scenes.

Moreover, [2] to facilitate rigorous evaluation and benchmarking of abnormal activity detection methods, we introduce a comprehensive dataset specifically curated for this task. [6] This dataset encompasses a diverse range of abnormal activities and scenarios, ensuring the robustness and generalization capability of the proposed framework.

In this paper, we present the design and implementation of the LRCN framework for abnormal activity detection. We conduct extensive experiments on benchmark datasets, demonstrating the efficacy and superiority of our proposed approach compared to existing methods [9]. Furthermore, we provide insights into the interpretability of the learned features and discuss potential applications and future directions for research in abnormal activity detection using deep learning techniques. Overall, [4] this work contributes to advancing the state-of-the-art in safety and security systems by providing more reliable and efficient mechanisms for abnormal activity detection.

II. BACKGROUND THEORY

In this section, we discuss concept of LRCN and comparison of different neural network architectures.

A. Concept of LRCN

Long-term Recurrent Convolutional Networks (LRCN) represent a fusion of Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs), specifically Long Short-Term Memory (LSTM) networks [5]. LRCN addresses the need for comprehensive analysis of both spatial and temporal information in video sequences, a requirement crucial for tasks such as abnormal activity detection. CNNs excel at extracting spatial features from images through convolutional layers and pooling operations, making them well-suited for tasks like image classification and object detection. On the other hand [5], RNNs, including LSTM networks, are designed to model sequential data by capturing temporal dependencies, proving effective in tasks such as natural language processing, time series analysis, and video processing.

LRCN combines the strengths of CNNs and LSTMs to jointly model spatial and temporal information in video sequences. It extends upon the concept of Convolutional LSTM (ConvLSTM) by incorporating convolutional layers before LSTM layers. [2] This integration allows LRCN to capture both short-term spatial features and long-term temporal

dependencies within a single framework. By leveraging both spatial and temporal information simultaneously, [13] LRCN offers a holistic approach to video analysis, addressing the limitations of traditional CNNs and RNNs in capturing comprehensive understanding of video data.

B. Comparison of Different Neural Network Architectures

Various neural network architectures have been proposed for video analysis tasks, each with its strengths and limitations. [2] Traditional CNNs are effective in extracting spatial features from images, making them suitable for tasks like image classification and object detection. However, they may struggle with capturing temporal dependencies in sequential data such as videos and may be limited in understanding long-term context. [7] Similarly, traditional RNNs are proficient in modeling temporal dependencies but are prone to vanishing gradients and may not effectively capture spatial features.

In contrast, LRCN integrates both spatial and temporal information seamlessly, capturing short-term spatial features and long-term temporal dependencies. This holistic approach to video analysis makes LRCN [5] robust to variations and complexities in video data, suitable for tasks requiring comprehensive understanding of video sequences such as abnormal activity detection. By outperforming traditional CNNs and [13] RNNs in these tasks, LRCN showcases its efficacy in real-world applications and underscores the importance of integrating spatial and temporal information in video analysis.

TABLE 1: COMPARISON OF DIFFERENT TYPES OF NEURAL NETWORK

Aspect	LRCN	CNNs	RNNs
Spatial Feature Extraction	Utilizes CNNs	Excels in spatial feature extraction	Limited in capturing spatial features
Temporal Modeling	Incorporates LSTMs	Focused on spatial feature extraction	Proficient in capturing temporal dependencies
Handling Sequential Data	Integrates spatial and temporal information	Not explicitly designed for sequential data	Specifically designed for sequential data
Long-term Dependency Handling	Addresses challenges of long-term dependencies	Limited in handling long-term dependencies	Prone to vanishing gradients and long-term dependencies
Application in Video Analysis Tasks	Well-suited for comprehensive video understanding	Suitable for image-related tasks	Effective in time series analysis and natural language processing
Robustness to Variations and Complexities	Robust to variations and complexities in video data	May struggle with variations and complexities	Sensitive to variations and may require careful handling
Performance in Abnormal Activity Detection	Outperforms traditional architectures	Limited capability in comprehensive video understanding	May lack robustness and accuracy in abnormal activity detection tasks

III. ISSUES INVOLVED IN ABNORMAL ACTIVITY DETECTION

Abnormal activity detection encompasses various challenges stemming from the complexity and variability inherent in real-world scenarios. Firstly, defining anomalies can be subjective and context-dependent, leading to ambiguity in labeling and potentially biased model training. Moreover, datasets often suffer from class imbalance, where normal activities vastly outnumber abnormal ones, posing challenges in training models that can generalize well. Additionally, [11] anomalies exhibit diverse temporal and spatial characteristics, requiring detection systems to accommodate variations in speed, scale, duration, and context across different scenarios. Noise and background clutter further complicate detection, as distinguishing true anomalies from benign variations or noise is challenging. Real-time processing is essential for applications like video surveillance, necessitating efficient algorithms with minimal latency. [4] Furthermore, ensuring models generalize across domains is crucial for practical deployment, requiring robustness and adaptability to diverse settings. Ethical and privacy concerns also loom large, especially regarding data collection, storage, and potential misuse of surveillance technologies, necessitating adherence to legal regulations and ethical standards. Addressing these challenges demands interdisciplinary collaboration, innovative algorithms, and rigorous evaluation methodologies to develop effective and reliable abnormal activity detection systems capable of meeting diverse application needs while upholding ethical and privacy considerations.

1) Dataset

The [12] KTH dataset stands as a cornerstone in the field of action recognition, renowned for its comprehensive collection of sequences representing six distinct actions. Within each action class, [11] the dataset boasts a robust repository of precisely 100 sequences, ensuring ample data for thorough analysis and model training. Each sequence within the dataset is meticulously crafted, comprising nearly 600 frames meticulously captured at a consistent frame rate of 25 frames per second (fps).

This meticulous curation of sequences and frames offers researchers and practitioners in the field of computer vision and machine learning a rich resource for studying and developing algorithms for action recognition tasks. [5] The diversity of actions covered in the dataset, coupled with the large number of sequences per class, allows for robust evaluation and [12] benchmarking of algorithms across various action categories.

2) Data Pre-processing

To preprocess the videos for action recognition, we employ a systematic approach using the [15] OpenCV library. Firstly, videos are read from their corresponding class folders, with each video labeled according to its class. This information is then stored in a numpy array, allowing for efficient handling and organization of the data.

Subsequently, each video undergoes a crucial step of splitting into frames to create sequences suitable for analysis. [11] Using OpenCV, we extract 30 frames from the video at equal time intervals, ensuring that each sequence contains a consistent representation of the action captured. This segmentation process enables the creation of standardized sequences for further processing.

Following [14] frame extraction, image resizing becomes imperative to maintain uniformity in input dimensions across all frames. Resizing each frame to a standardized width and height of 64 pixels ensures consistency in input size, facilitating streamlined processing and analysis.

Normalization of the resized frames follows suit, serving to expedite the learning process of the subsequent model. By dividing the pixel values of each frame by 255, we scale the pixel intensities to fall within the range of 0 to 1. [1] This normalization step aids the learning algorithm in capturing essential features from the images more effectively, thereby enhancing the model's performance.

Finally, the processed frames are consolidated into numpy arrays, forming sequences comprising 30 resized and normalized frames.[6] These arrays serve as input data for the action recognition model, providing a structured and standardized representation of the video sequences. [10] This meticulous preprocessing pipeline ensures that the input data is appropriately formatted, standardized, and optimized for effective model training and evaluation.

After preprocessing, a vital train-test split stage is performed on the dataset to support effective model training [13] and evaluation. This involves dividing the dataset into training and testing subsets to thoroughly assess the model's performance.

The [5] dataset is divided into training and testing sets with precision, allocating 75% of the data for training and reserving the remaining 25% for testing. This meticulous partitioning ensures ample data for the model to learn underlying patterns and features during training, while also offering an independent dataset for evaluating the model's generalization and predictive abilities.

3) Model Creation

In our proposed system for suspicious activity detection from video surveillance, we employ a deep learning network known as Long-term Recurrent Convolutional Network (LRCN) [11]. This architecture was inspired by a class of end-to-end trainable architectures for visual recognition and description, which was introduced by a group of authors in 2016. The fundamental concept behind LRCN is to combine Convolutional Neural Networks (CNNs) for learning visual features from video frames with [9] Long Short-Term Memory (LSTM) networks for transforming a sequence of image embeddings into various outputs such as class labels, sentences, or probabilities.

CNNs are adept at extracting spatial features from images, making them well-suited for tasks like image classification and object detection. [7] By processing raw visual input with a CNN, we can effectively capture relevant visual information from video frames. The outputs of the CNN are then fed into a stack of LSTM layers, which are specialized recurrent neural networks designed to handle sequential data.

LSTM networks are particularly well-suited for tasks involving time series data, as they can effectively model temporal dependencies and capture patterns with lags of unknown duration between important events. One of the key advantages of [11] LSTMs over traditional Recurrent Neural Networks (RNNs) is their ability to mitigate the vanishing gradient problem, which arises during the training of RNNs and can impede the learning process over long sequences.

In the context of suspicious activity detection from video surveillance, [13] LRCN leverages the complementary strengths of CNNs and LSTMs. The CNN component extracts visual features from video frames, while the LSTM component processes these features over time, enabling the model to analyze temporal dynamics and detect anomalous behavior patterns. [15] By combining these two architectures in an end-to-end trainable manner, LRCN offers a powerful framework for capturing complex spatiotemporal relationships in video data and detecting suspicious activities with high accuracy and efficiency.

4) Model Training

In our model designed for suspicious activity detection, the training task revolves around predicting one of three distinct classes: walking, running, or fighting. To train the model effectively, [2] we employ a set of carefully selected hyperparameters, which are crucial for optimizing the learning process and achieving desirable performance.

Firstly, [9] we set the number of epochs, which dictates the number of complete passes through the entire training dataset during the training process. In this scenario, we opt for a total of 70 epochs, allowing the model to iteratively learn from the training data over multiple cycles.

Additionally, we specify the batch size, which determines the number of samples processed by the model in each training iteration. A batch size of 4 is chosen for this training regimen, enabling efficient utilization of computational resources while [10] ensuring stable and consistent updates to the model's parameters.

Furthermore, we incorporate a validation split parameter, which defines the proportion of the training data reserved for validation purposes during the training process. Here, a validation split of 0.25 (equivalent to 25% of the training data) is selected, [15] indicating that a quarter of the training dataset is set aside for validation to monitor the model's performance and prevent overfitting.

By configuring these hyperparameters, we aim to strike a balance between model complexity, training efficiency, and generalization capability. [8] Through meticulous tuning and optimization of these parameters, we endeavor to train a robust and accurate model capable of effectively classifying suspicious activities from video surveillance data with high precision and reliability.

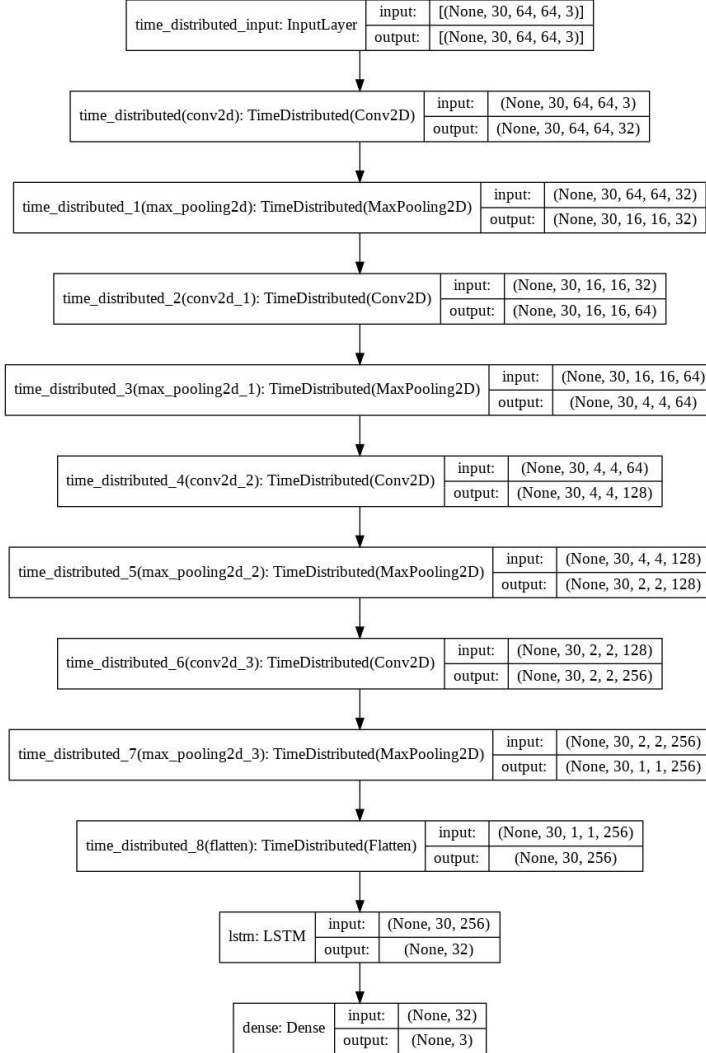


FIG 1: PROPOSED LRCN MODEL

5) Results

Our custom Long-term Recurrent Convolutional Network [11] (LRCN) model, comprising 12 layers, underwent training and evaluation using a dataset featuring 100 videos per class. To optimize efficiency on various devices, frames within the dataset were resized to 64 x 64 pixels. This resizing ensured swift data loading, [4] even on low-end devices, while preserving adequate visual detail for precise detection. [9] Such customization was geared towards real-life scenarios, prioritizing quick prediction times and suitability for deployment on resource-constrained platforms.

In comparison to the benchmark [3] VGG-16 + LSTM model, which processed 45 videos per class at a frame size of 224 x 224 pixels, our LRCN model demonstrated competitive performance with an accuracy of 82%. Notably, [8] our model achieved near real-time prediction capabilities, ensuring swift response times suitable for real-world surveillance applications.

The following table summarizes the performance comparison between the VGG-16 + LSTM model and our custom LRCN model:

TABLE 2: RESULTS FOR PROPOSED LRCN MODEL

Model	Dataset	Frame Size	Accuracy	Near Real-time Prediction
VGG-16 + LSTM	45 videos/Class	224 x 224px	85%	No
LRCN (Custom)	100 videos/Class	64 x 64px	82%	Yes

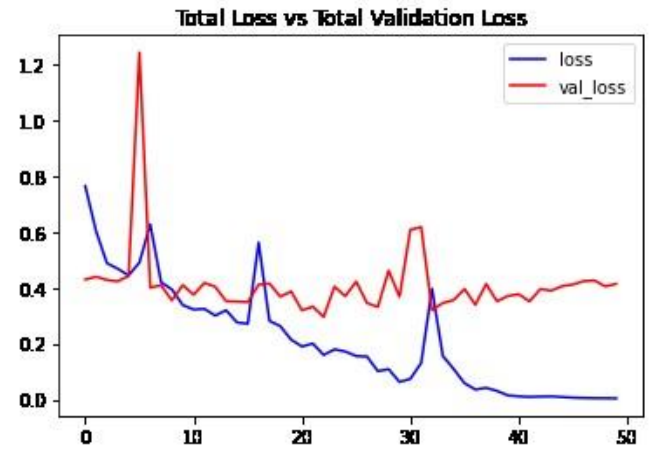


FIG 2: LOSS VS VALIDATION LOSS

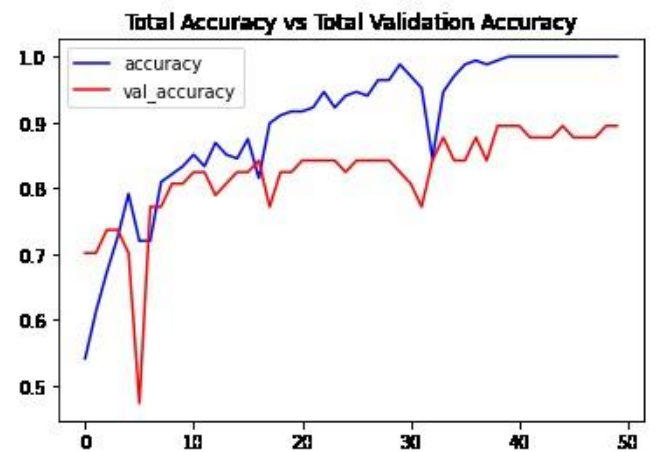


FIG 3: ACCURACY VS VALIDATION ACCURACY

IV. LITERATURE SURVEY

The paper discusses the use of CCTV cameras for surveillance in public places and the need for detecting abnormal activities in real-time. It highlights the challenges in detecting abnormal crowd behavior, such as climate change, human movement, occlusions, and low video quality. The paper explores the use of deep learning and handcrafted approaches for abnormal activity detection [1]. It provides an overview of various methods and techniques used in abnormal activity detection, including dynamic abnormal detection mechanisms. The paper emphasizes the high dimensionality of video data and the increased space and time complexity in abnormal activity detection.

The paper focuses on developing a novel system for automated video surveillance that can detect abnormal activities such as road accidents, falls, fires, and weapons. The fall detection algorithm is designed to detect posture-grounded events in a senior monitoring operation in a home surveillance system. [2] The fire detection algorithm is efficient and can automatically detect fire disasters in the surveillance area through real-time analysis of recorded videos. The auto crash detection system helps in recycling surveillance videos by directly sending them to emergency services without any intervention, reducing communication latency and saving lives during accidents. The weapon detection system aims to reduce violence by identifying weapons in surveillance videos.

The paper proposes an unsupervised method for detecting anomalous sounds in machine condition monitoring by utilizing an auxiliary task of detecting machine activity [3]. The method involves training a model using normal data with machine activity labels to detect machine activity. The activity-detection error is used as the anomaly score for a given sound clip, either through ground-truth activity labels or outlier detection on embedding vectors. The proposed method outperforms conventional methods, especially in environments with similar background noise to the target machine sound. The method also improves anomaly detection performance through ensemble techniques.

The paper focuses on the development of a deep learning-based algorithm for detecting abnormal activity in surveillance systems. It introduces two approaches: a spatiotemporal auto-encoder based on a 3D convolution neural network, and a pre-trained model using C3D feature extraction. The first approach, the spatiotemporal auto-encoder, achieves a minimum loss of 8% and a maximum accuracy of 71% in reconstructing frames. The second approach, C3D feature extraction, improves accuracy by computing features from the fully connected layers of the C3D model. The proposed algorithm can be implemented through CCTV cameras in various crowded areas such as banks and roads to automate and simplify manual work. The paper highlights the importance of abnormal activity detection in real-world scenarios, including accidents, burglary, explosion, fighting, robbery, and other critical events. [4] The use of deep learning techniques, such as auto-encoders and convolutional neural networks, shows promise

in reducing manual work and time required for abnormal activity detection.

[5] The paper proposes a method for abnormal activity detection based on the MCRF (Multiple Conditional Random Fields) model. It highlights the advantage of the MCRF model in combining multiple features of abnormal activity and utilizing adaptive contextual information. The method involves extracting several feature subsets and applying the CRF (Conditional Random Fields) model to each subset to obtain CRF units. These CRF units are then combined to form the MCRF model, which is used for detecting abnormal activity. The experimental results indicate that the proposed method achieves better detection accuracy compared to other methods. Additionally, the use of parallel processing reduces the time required for model training.

The paper proposes a method for abnormal activity detection in indoor environments using radio tomographic networks. The method does not require computed recovery imaging and instead encodes the critical state feature of abnormal activity into the data stream of received signal strengths (RSSs). [6] The proposed method compares the new sensor data with the recorded instantaneous state feature to perform abnormal detection based on similarity. The efficacy of the method is validated through experiments, where walking is considered normal activity and falls are considered abnormal activity. The results of the experiments are encouraging, indicating the effectiveness of the proposed method for abnormal activity detection.

The paper focuses on the detection and localization of human anomalous activities inside the ATM room using a computer vision-based system. [7] The authors highlight the increase in criminal activities related to ATM installations, such as attacks on customers and snatching of money, which affect customer security. The proposed system aims to monitor human behaviors by learning patterns of normal and abnormal activities in ATM surveillance videos. The system achieves better accuracy with less error rate when varying the bin size, as demonstrated through experiments comparing it with existing methods.

The paper proposes a multi-sensing modality framework for human activity recognition and abnormal behavior detection by combining skeleton and acceleration data at the feature level. The framework utilizes deep temporal convolutional networks (TCN) with dilated causal convolution components for feature learning and handling temporal properties. Experimental results show that the proposed method achieves an 83% F1-score on the Continuous Multimodal Multi-view Dataset of Human Fall Dataset (CMD FALL) and 96.98% accuracy on the UTD-MHAD dataset. [8] The proposed method outperforms several single modality models and early and late fusion methods, indicating its effectiveness for fine-grained activity recognition and abnormal behavior detection. The results also suggest the potential of the proposed method for HAR-based situated service applications.

The paper proposes a novel method for anomaly detection in the daily routine and behavior of elderly people using a ConvLSTM Autoencoder. [9] The authors validate their

proposed method by comparing it to other state-of-the-art approaches using two datasets from the WSU CASAS smart home project. The paper highlights that anomalies in elderly behavior are sparse and rare, making them harder to recreate in an Autoencoder and resulting in a higher reproduction error compared to normal behavior. The authors of the paper "One of the most successful Anomaly Detection methods is One-Class Support Vector Machine (OCSVM)" compare their results with this algorithm and show significant improvement in the context of Ambient Intelligence. The paper utilizes ConvLSTMAE to process spatiotemporal data and demonstrates tremendous improvement over state-of-the-art methods in detecting abnormal behavior related to dementia in elderly people in smart homes. The authors mention that Recurrent Neural Networks (RNNs) are used to process sequential data and model temporal dynamic behavior, which is important in activity detection.

[10] The paper addresses the issue of abnormal activity detection in surveillance videos, which has gained significant importance in security and surveillance applications. The use of CCTV cameras for video recording in public spaces has become common, but the reliance on human monitors for post-incident manual analysis poses risks to security. The proposed separable convolution neural network aims to enable surveillance cameras to understand activities in real-time and report events that require attention, enhancing the reliability of the surveillance system. The study evaluates the performance of the proposed model using different optimizers, including Adam, RMSProp, and SGD. The evaluation metrics used include confusion matrix, accuracy, classification report with sensitivity, specificity, precision, and f1-score. The results show that the separable convolution neural network model trained with SGD optimizer achieves 94% accuracy with the test data.

Obstructive Sleep Apnea (OSA) is a common condition that causes sleep disturbance and daytime sleepiness, with potential serious consequences for individuals and society. [11] The severity of OSA is determined by the frequency of apnea episodes, where breathing stops due to the constriction of the throat during sleep. Various parameters, such as EEG sleep staging, snoring, changes in airway resistance, airflow, inspiratory effort, oxygen saturation, and body movement, are used to understand the pathophysiology of sleep apnea. Nocturnal oxygen saturation and movement during sleep are the best predictors of morbidity in OSA patients, as assessed by improvements with continuous positive airway pressure therapy (CPAP). Video monitoring has been adopted to assist in the diagnosis of OSA, but there are challenges in video analysis due to lighting restrictions, body obscuration, variation in human size and behavior, and the volume of video and audio data. Automated monitoring of human breathing activity is needed to diagnose sleep disorders more efficiently.

The paper proposes a method to detect abnormal events in the daily life of solitary elders by mining data gained from sensors. The study analyzes the relationship between the distribution of an elder's activity time and their activity and obtains association rules for the elder's actions. [12] The goal

of the research is to detect abnormal activity and provide timely and appropriate services to solitary elders.

The paper presents a method for abnormal activity detection in intelligent home care, specifically focusing on detecting falling down as an abnormal activity. [13] The activities are categorized into six classes: standing, sitting, squatting, walking, jogging, and falling down. Falling down is considered abnormal, including on-marching falling down and in-place falling down. The paper uses background subtraction to obtain binary human-body images and selects centroid track and figure width of human blob as features for recognition. The activities are further divided into moving activities and quasi-static activities based on the horizontal movement of the centroid of the body blob. SVM classifiers are then used to recognize on-marching falling down and in-place falling down. The experimental results show a correct identification rate of 100% for on-marching falling down and a minimum correct identification rate above 90% for in-place falling down activity.

The paper presents a novel two-phase approach for detecting abnormal activities based on wireless sensors attached to a human body. Traditional approaches to detecting abnormal activities suffer from a high false positive rate, especially when the collected sensor data are biased towards normal data while abnormal events are rare. [14] The proposed approach first employs a one-class support vector machine (SVM) trained on commonly available normal activities to filter out activities that are likely to be normal. Abnormal activity models are then derived from a general normal model using kernel nonlinear regression (KNLR) to reduce the false positive rate in an unsupervised manner. The approach provides a good tradeoff between abnormality detection rate and false alarm rate and allows for the automatic derivation of abnormal activity models without the need for explicitly labeled abnormal training data. The effectiveness of the approach is demonstrated using real data collected from a sensor network deployed in a realistic setting. Activity-based abnormality detection techniques typically compare a profile of all normal activities to actual observations, flagging any deviation from the profile as a potential alarm. The positive activity model assumes that everything other than the identified normal activities is abnormal. Previous work has used switching hidden semi-Markov models to represent user's activities and perform abnormality detection, focusing on abnormalities in state duration rather than state order.

[15] The paper introduces a novel abnormal activity detection algorithm for multi-camera surveillance. It represents the scene covered by the cameras as a network, with nodes representing scene segments and edges denoting activity correlations. Human activities are modeled as signal transmissions in this network, facilitating abnormal activity detection based on "network transmission energy". The algorithm is versatile, accommodating various multi-camera setups, making it more generalizable than prior methods. Experimental results attest to its effectiveness, offering improved accuracy and flexibility in abnormal activity detection across multiple cameras.

TABLE 3: LITERATURE SURVEY TABLE

Citation & Year	Objective	Classifier	Evaluation Metrics	Results
[1], 2023	Offers comprehensive survey on abnormal activity detection techniques in intelligent video surveillance.	CNN, RNN, SVM, RF	Precision, recall, F1 score, accuracy	Abnormal activity detection in video surveillance using deep learning and handcrafted features.
[2], 2023	Create innovative automated video surveillance system to detect and identify abnormal activities in CCTV recordings.	CNN	F1 score, and area under the receiver operating characteristic curve	Innovative system automates video surveillance, detects road accidents, falls, fires, and weapons.
[3], 2022	Create unsupervised anomalous sound detection for machine condition monitoring, leveraging auxiliary task of identifying machine activity.	CNN	Precision, recall, F1 score	Proposed method enhances anomaly-detection performance compared to conventional approaches.
[4], 2022	Enhance accuracy, reduce human workload in identifying critical events like accidents or robberies.	CNN	Recall, F1 score, accuracy	Designed for crowded areas, like banks and roads, utilizes two deep learning approaches for high accuracy, compatible with existing CCTV cameras.
[5], 2018	Authors propose MCRF model for more accurate abnormal activity detection, integrating diverse features and adaptive contextual information.	CRF	F1 score	Method employs MCRF model to integrate diverse features and contextual information for improved accuracy.
[6], 2017	Propose abnormal activity detection using radio tomographic network RSSs.	HMM	F1 Score	Designed vertical profile-aware network for abnormal activity detection in radio tomographic networks using RSS data.
[7], 2022	Proposes a computer vision-based system for recognizing abnormal activities inside ATM rooms.	CNN	Precision, recall, F1 score	Designed computer vision system for ATM room abnormal activity detection.
[8], 2021	Combine skeleton & acceleration data for fine-grained activity recognition & abnormal behavior detection.	TCN, Deep TCN	F1-score	Achieves 83% F1-score on CMDFALL & 96.98% accuracy on UTD-MHAD dataset.
[9], 2021	Propose ConvLSTM Autoencoder for detecting abnormal elderly behavior in routines.	ConvLSTM Autoencoder, CNN, RNN, LSTM	Precision, Recall, Specificity, F-score, ROC, AUC	ConvLSTM Autoencoder outperforms state-of-the-art in elderly abnormal behavior detection.
[10], 2021	Implement Separable CNNs for detecting abnormal activities in surveillance videos.	SCNN	ROC, AUC, Precision, Recall, F1 Score, FPR	Explored SCNN for detecting unusual activities in surveillance videos.
[11], 2008	Study proposes & evaluates SCNN for classifying surveillance video activities.	SCNN	Confusion Matrix, Accuracy, Classification Report	SCNN with SGD optimizer attains 94% accuracy on test data.
[12], 2004	Paper aims to detect abnormal events in solitary elders' daily lives using sensor data.	CNN	Recall, F1 score, accuracy	Analyzes elder activity timing, deriving association rules for abnormal behavior detection, enabling timely assistance..
[13], 2008	Create home care abnormal activity detection using SVM and background subtraction.	SVM	Correct identification rate	Achieves 100% correct identification for on-marching falling and over 90% for in-place falling activity.

[14], 2008	Introduces novel two-phase approach for abnormal activity detection with body-attached wireless sensors.	One-Class SVM	Detection Rate, False Alarm Rate	Utilizes one-class SVM trained on normal activities to filter out probable normal activities.
[15], 2011	Proposes Network-Transmission-Based (NTB) algorithm for multi-camera surveillance abnormal activity detection.	NTB, TSB	Miss, TFER	NTB algorithm surpasses TSB in miss and Total Error Frame Rate.

V. CONCLUSION

In conclusion, our paper introduces a robust approach for suspicious activity detection using a [11] Long-term Recurrent Convolutional Network (LRCN), effectively classifying activities into walking, running, and fighting categories. Through the fusion of Convolutional Neural Networks (CNNs) for spatial feature extraction and Long Short-Term Memory [14] (LSTM) networks for temporal modeling, our framework achieves accurate detection of anomalous behavior patterns in video surveillance data. By fine-tuning hyperparameters such as epochs, batch size, and validation split, we ensure efficient model training and optimal performance.

Future work will focus on enhancing the scalability and adaptability of our approach to handle a broader range of anomalous behaviors and environmental conditions. Exploring advanced techniques such as attention mechanisms and multimodal fusion could further improve detection accuracy and robustness. [15] Additionally, investigating real-time deployment strategies and integrating contextual information from other sensors could enhance the practical utility of the system in diverse surveillance scenarios.

REFERENCES

- [1] Priya, S. S., & Minu, R. I. (2023). Abnormal Activity Detection Techniques in Intelligent Video Surveillance: A Survey. In 2023 7th International Conference on Trends in Electronics and Informatics (ICOEI) (pp. 1608-1613). Tirunelveli, India. doi: 10.1109/ICOEI56765.2023.10125671.
- [2] Sittalatchoumy, R., Pushpa, S. E. P., Karthick, R., Azharudheen, M. A. R., & Dhiyanesh, K. V. (2023). Transfer Learning based Video Surveillance for Abnormal Activity Detection. In 2023 7th International Conference on Intelligent Computing and Control Systems (ICICCS) (pp. 419-423). Madurai, India. doi: 10.1109/ICICCS56967.2023.10142535.
- [3] Nishida, T., Dohi, K., Endo, T., Yamamoto, M., & Kawaguchi, Y. (2022). Anomalous Sound Detection Based on Machine Activity Detection. In 2022 30th European Signal Processing Conference (EUSIPCO) (pp. 269-273). Belgrade, Serbia. doi: 10.23919/EUSIPCO55093.2022.9909901.
- [4] Dhanush Kumar, A., Shushruth Reddy, P., Parikh, K. C., Meghana Sarvani, C., & Loel Maansi, P. (2022). Abnormal Activity Detection Using Deep Learning. In A. K. Nagar, D. S. Jat, G. Marín-Raventós, & D. K. Mishra (Eds.), Intelligent Sustainable Systems (pp. 333). Lecture Notes in Networks and Systems. Springer, Singapore. doi: 10.1007/978-981-16-6309-3_63.
- [5] Long, Z., Bo, G., Guoqiang, Z., Xin, W., Rujia, Q., & Wenjun, X. (2018). Abnormal Activity Detection based on MCRF. In 2018 2nd IEEE Conference on Energy Internet and Energy System Integration (EI2) (pp. 1-5). Beijing, China. doi: 10.1109/EI2.2018.8582147.
- [6] Liu, T., Wei, X. -h., Chen, Z. -m., Luo, X. -m., & Liu, J. (2017). Abnormal activity detection based on received signal strengths of radio tomographic networks. In 2017 29th Chinese Control And Decision Conference (CCDC) (pp. 7310-7314). Chongqing, China. doi: 10.1109/CCDC.2017.7978505.
- [7] Prabha, B., Manivannan, P., & Nagesh, P. (2022). Human Abnormal Activity Detection in the ATM Surveillance Video. In P. S. R. Chowdary, J. Anguera, S. C. Satapathy, & V. Bhateja (Eds.), Evolution in Signal Processing and Telecommunication Networks (pp. 5). Lecture Notes in Electrical Engineering. Springer, Singapore. doi: 10.1007/978-981-16-8554-5_5.
- [8] Pham, C., Nguyen, L., Nguyen, A., et al. (2021). Combining skeleton and accelerometer data for human fine-grained activity recognition and abnormal behaviour detection with deep temporal convolutional networks. Multimedia Tools and Applications, 80, 28919-28940. https://doi.org/10.1007/s11042-021-11058-w
- [9] Nazerfard, E., Atashgahy, Z., Nadali, A., et al. (2021). Abnormal Activity Detection for the Elderly People using ConvLSTM Autoencoder. Preprint available at Research Square. https://doi.org/10.21203/rs.3.rs-693084/v1
- [10] Anju, S. S., & Kavitha, K. V. (2021). Separable Convolution Neural Network for Abnormal Activity Detection in Surveillance Videos. In J. S. Raj, A. M. Iliyasu, R. Bestak, & Z. A. Baig (Eds.), Innovative Data Communication Technologies and Application (pp. 29). Lecture Notes on Data Engineering and Communications Technologies, 59. Springer, Singapore. https://doi.org/10.1007/978-981-15-9651-3_29
- [11] Wang, C. W., et al. (2008). The Detection of Abnormal Breathing Activity by Vision Analysis in Application to Diagnosis of Obstructive Sleep Apnea. In N. Wickramasinghe & E. Geisler (Eds.), Encyclopedia of Healthcare Information Systems (pp. 416-424). IGI Global. https://doi.org/10.4018/978-1-59904-889-5.ch053
- [12] Tang, Y., Zheng, Z., Gu, H., Lu, H., Wang, W., & Xiao, G. (2004). Elder Abnormal Activity Detection by Data Mining. SICE Annual Conference Program and Abstracts. https://doi.org/10.11499/sicep.2004.0.52.1
- [13] Qian, H. M., Mao, Y. B., & Wang, Z. Q. (2008). SVM-based abnormal activity detection for home care. In 2008 7th World Congress on Intelligent Control and Automation (pp. 3766-3771). Chongqing, China. https://doi.org/10.1109/WCICA.2008.4593529
- [14] Yin, J., Yang, Q., & Pan, J. J. (2008). Sensor-Based Abnormal Human-Activity Detection. IEEE Transactions on Knowledge and Data Engineering, 20(8), 1082-1090. https://doi.org/10.1109/TKDE.2007.1042
- [15] Gao, H., et al. (2011). A new network-based algorithm for multi-camera abnormal activity detection. In 2011 IEEE International Symposium of Circuits and Systems (ISCAS) (pp. 361-364). Rio de Janeiro, Brazil. https://doi.org/10.1109/ISCAS.2011.5937577.

